

Tribhuvan University
Institute of Science and Technology
SCHOOL OF MATHEMATICAL SCIENCES
Second Reassessment 2083

Subject: Regression Analysis
Course No: MDS 605
Level: MDS / II Year / III Semester

Full marks: 45
Pass Marks: 22.5
Time: 2hrs

Candidates are required to give their answers in their own words as far as practicable.
Attempt All Questions.

Group A

[5x3=15]

1. Explain BLUE properties briefly.
2. The following regression output is obtained from R

Variable	Estimate	Std. Error	t-value	p-value
Intercept	15.32	2.18	7.03	<0.001
Income	0.48	0.09	5.33	<0.001
Age	-0.12	0.05	-2.40	0.020
Education	1.10	0.37	2.97	0.005

- a) Write the fitted regression equation.
- b) Which predictors are statistically significant?
- c) Interpret the coefficient of Income and Education.

3. The software provides the following output:

Variable	VIF
X_1	2.45
X_2	11.62
X_3	3.28

1-5
5-10
10-20

Which variable shows evidence of multicollinearity? Suggest technique to reduce multicollinearity.

4. A researcher estimates the following model to examine the effect of education and gender on annual income:

$$\text{Income} = 120000 + 18000(\text{Education}) + 25000(\text{Male})$$

where

Education = Years of education

Male = 1 if male, 0 if female

- a) Interpret the coefficient of the dummy variable.
- b) Predict the annual income of:

- A male with 16 years of education.
A female with 16 years of education.
- c) What is the expected income difference between a male and a female with the same education level?

5. Differentiate among Forward Selection, Backward Elimination, and Stepwise Regression.

Group B [5×6= 30]

6. Explain model specification and assumptions of multiple regression analysis.
7. Explain the consequences of autocorrelation on the assumptions and properties of the Ordinary Least Squares (OLS) estimators. Discuss the remedial measures that can be adopted to correct the problem of autocorrelation in econometric models.

Or

A researcher estimates a consumption function using annual data from 2011–2025 (15 years). The following output is obtained:

$$R^2 = 0.93$$

$$F = 105.64 \text{ (} p < 0.001 \text{)}$$

$$\text{Durbin-Watson} = 0.71$$

$$\text{Breusch-Godfrey test } p\text{-value} = 0.0003$$

- Explain whether the model suffers from autocorrelation.
- Compare the conclusions from the Durbin-Watson and Breusch-Godfrey tests.
- Explain how autocorrelation affects the efficiency of OLS estimators.
- Describe two econometric methods that could be used to correct the problem.

8. Define multicollinearity and distinguish between perfect and imperfect multicollinearity. Describe the methods used to detect multicollinearity.

9. Discuss the methods used to detect heteroskedasticity in a regression model.

p

Describe Goldfeld -Quandt Test to test the heteroskedasticity briefly.

10. Define logistic regression with its assumptions. Also explain Hosmer Lemeshow test and ROC curve.
